

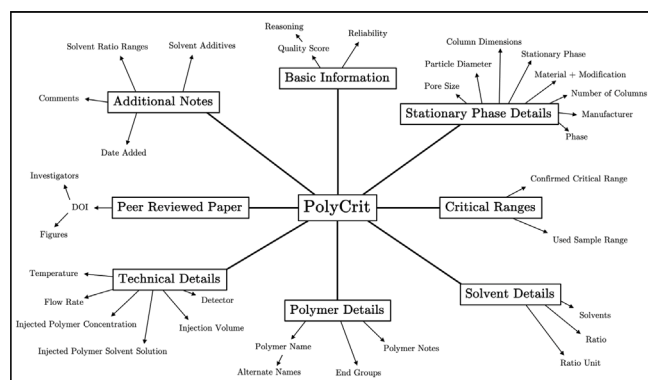


Fig. 2. Cluster diagram detailing different tracked parameters and their respective categories. Arrows between parameters indicates flow of creation.

The partition coefficient K , influenced by the ratio of the polymer's radius of gyration (R , a measure of M_w) to the pore size (D), determines how polymers interact with the stationary phase. In Eq. (1), c is the adsorption interaction parameter used to measure the potential interaction energy between the polymer and the stationary phase. In SEC, $c < 0$, indicating that the interaction energy of the polymer is weaker than the critical threshold, resulting in elution based primarily on molecular weight (M_w). Conversely, in LAC, $c > 0$, signifying stronger polymer interactions with the stationary phase, leading to adsorption. At the CAP, $c = 0$ and thus K approaches 1, independent of pore size or chain length, reflecting a balance where adsorption neither dominates nor depletes polymer density near the wall [8]. However, in practice, seeking conditions for $K = 1$ has been shown to not be particularly viable in identifying LCCC [6,9–11]. Instead, experimentalists typically will simply identify the phenomenon based on its characteristic vertical line (as in Fig. 1).

3. Materials and methods

In developing this database, we undertook an extensive literature review to compile LCCC data spanning several decades. This review included studies from as early as 1976 up to the most recent publications in 2023, reflecting the evolution and refinement of LCCC methodologies over time. Particularly, we searched through major chromatographic journals picking papers mentioning relevant key terms also found in this paper. Review papers written by Macko et al. [12] in 2003 and Baumgaertel et al. [13] in 2012, as well as an early database paper by Trathnigg et al. [14] in 2007 were of great help during this review. Their works inspired the creation of this database platform, and their precedents helped us decide what experimental conditions needed to be tracked.

The selection process was rigorous; we focused on studies that provided detailed chromatographic conditions, including polymer type, solvent combinations, stationary phases, and other relevant parameters (a full list can be found in Fig. 2). Each paper was carefully examined to verify that critical conditions were accurately achieved and documented. In total, we reviewed 312 papers on the topic, recorded 625 conditions, and selected 428 unique LCCC conditions, each validated and curated for inclusion in the database.

3.1. Curation

Some entries were initially reported by other publications as being critical but upon further review did not present sufficient evidence and were removed from the dataset. Additionally, similar systems were present in the literature multiple times. Such entries were combined at the discretion of the reviewer into one entry. If some conditions were

Table 1

An example of data curation by consolidating similar entries where the top and bottom tables are examples of the before and after.

Polymer	Solvent	Solvent ratio
Polystyrene	DCM, ACN	58.0, 42.0
Polystyrene	DCM, ACN	58.2, 41.8

Polymer	Solvent	Solvent ratio	Solvent ratio ranges
Polystyrene	DCM, ACN	58.1, 41.9	58.0–58.2% DCM

Table 2

Reliability scores for assessing the independence of molecular weight (M_w) and retention time in critical condition data.

Reliability	Reasoning
Excellent	5 or more data points clearly indicating an independent (vertical) relationship between M_w and retention time.
Good	4 or more data points roughly indicating an independent relationship, but not obviously vertical.
Low	2 to 3 data points roughly indicating an independent relationship, but not obviously vertical.
Indirect	Critical conditions stated, but no evidence given.

similar enough to be combined but not exact, a note was left to describe the range of the critical conditions (see Table 1 for an example).

To combine two entries, nearly all conditions had to match exactly, with allowances for differences in only one or two criteria. Uncombined records were retained for documentation purposes. After this thorough selection process, we turned our attention to assessing the quality of each condition, recognizing that the reliability of the data is essential for meaningful analysis and comparison.

3.2. Solvent ratio unit

Several critical conditions involved solvent ratios expressed in units of wt% or mol%. Given the potential for machine learning applications in this project, we decided to convert all these ratios to vol%, the most commonly used unit among the reviewed studies.

Solvent ratios were normalized to ensure consistency. This normalization step adjusts the ratios so that their sum equals 100%, calculated as:

$$\text{Normalized Ratio}_i = \frac{\text{Original Ratio}_i}{\sum_j \text{Original Ratio}_j} \times 100\% \quad (3)$$

where Original Ratio_i represents the proportion of the i th solvent in the mixture, and the denominator is the total of all solvent ratio components in that entry.

Next, solvent ratios were converted to volume units (vol%) for uniform representation. For ratios initially provided in weight (wt%), the conversion utilized the density (d) of each solvent:

$$\text{Volume Ratio}_i = \frac{\text{Weight Ratio}_i}{d_i} \quad (4)$$

For ratios provided in moles (mol%), the conversion accounted for both the molecular weight (M_w) and density (d) of each solvent:

$$\text{Volume Ratio}_i = \frac{\text{Mole Ratio}_i \cdot M_{w,i}}{d_i} \quad (5)$$

Finally, the ratios were again normalized to 100%. These calculations ensured that all solvent ratios were expressed in a comparable volumetric format.

3.3. Quality rating

By definition, the retention time and the M_w should have an independent relationship at LCCC (the vertical line shown in Fig. 1). However, since a continuous range of polymer weights cannot be evaluated, a relatively small number of vertically aligned data points